Journal
Travel to Bhubaneswar, India for the MPT-2009
J.D. Miller
24-30 October 2009

Saturday/Sunday 24, 25 October 2009

At the invitation of Dr. B.K.Mishra, Director of the National Laboratory and former student of Professor Rajamani of our Department, departure was made to participate in the International Seminar on Mineral Processing Technology, 26-30 October 2009 (MPT-2009) and give the Inaugural Lecture.

I was last in India in 1981 (my only previous visit) when Professor Herbst and I organized a two-week NSF Workshop for 21 US Scientists/Engineers dealing with the processing of mineral resources-including mineral processing, hydrometallurgy, and pyrometallurgy. One of our stops in 1981 was at the laboratory in Bhubaneswar at that time under the direction of Professor Jena. In addition to the visit at Bhubaneswar, we had stops at Hyderbad, Madras, Bombay, Agra (Taj Mahal) and Delhi before concluding with a week-long conference at Udaipur in Rajastan Province.

Although the travel is long, it will be interesting to visit India again after so many years have passed. Departure from SLC was 1:30 pm on Saturday, 24 October 2009, to Chicago and then from Chicago nonstop to Delhi arriving at 8:30 pm on Sunday, 25 October 2009. Both flights were on time and seating was good considering the flight to Delhi was 14 hours. Sleeping on the plane was difficult but thank God my back did not cause me too much discomfort. While changing planes in O'Hare I was able to see the end of the PSU/Michigan football game. Penn State won 35-10. I don't think the Nittany Lions had won in Ann Arbor since 1996. Also, before departure to Delhi, I was pleased to learn that the Utes had defeated Air Force in overtime. Both games were critical for these ranked teams to stay in contention for national recognition and participation in major bowl games at the end of the season. It was ironic that the flight plan for my 777 flight to Delhi took me right across the Michigan peninsula and Ann Arbor where PSU had just defeated the Wolverines.

Arrival in Delhi was without incident and I was received by a driver from the Radisson Hotel. All accommodations had been arranged by Dr. Mishra. The room was excellent and gave me some rest, although sound sleep was difficult, before departing for Bhubaneswar the morning of Monday, 26 October 2009. During the trip I prepared some Opening Remarks for the conference at the request of B.K.Mishra.

Opening Remarks MPT-2009

It is a special honor to be able to participate in the International Seminar on Mineral Processing Technology (MPT-2009) and to visit Bhubaneswar again. I was in Bhubaneswar in 1981 when Professor Herbst and I organized a two-week NSF Workshop for 21 US Scientists/Engineers dealing with the processing of mineral resources-including mineral processing, hydrometallurgy, and pyrometallurgy. One of our stops in 1981 was at this laboratory in Bhubaneswar, at that time under the direction of Professor Jena. My interest in India, particularly

Bhubaneswar, and its rich tradition in Mineral Processing Technology has been stimulated, in part, by many students who have studied with us at the University of Utah during the past 40 years, including, of course, B.K. Certainly it is my pleasure to meet with you for the International Seminar MPT-2009 conference.

We have made considerable progress since 1981, but many challenges still remain. The availability of mineral resources continues to be of international concern with demand for raw materials expected to grow significantly as the economies of China, India and other developing nations evolve to provide higher standards of living for their people. We must make more efficient use of our mineral and metal resources. This situation has a tremendous impact on our profession. The challenges and opportunities for mineral processing are many and include the matter of resources availability, technology development, including environmental considerations, and manpower. Every challenge represents an opportunity.

Mineral resources are sought around the world; in the Artic regions, in the South Sea Islands, in the Codilleras of South America, in Asia, and in the remote areas of Africa. Even in the U.S., new deposits are being discovered and, based on exploration, the extensions of old deposits are being established. For example, close to home in Utah, Rio Tinto has discovered what may be the world's largest deposit of moly, 500 million tons, 1500 feet below the floor of Kennecott's Bingham Canyon's open pit. AngloGold continues the development of gold operations at Cripple Creek in Colorado, mining an ore which has significant telluride mineralization. These are just two of many examples.

Existing technology for processing these and other mineral resources must be improved and new processing strategies developed. At least 30 years ago Professor Kellogg, from Columbia University, described the resource situation which defines our condition today, more than ever, we are continually technologically challenged and limited by the following,

Tyranny of Ore Grade
Tyranny of Ore Type

Today these mineral resource features represent the challenge we face. How will we economically process such ores that are of low grade or of difficult complexity? In some cases, the challenge may be met with improved design of equipment such as the first 300 m3 SuperCell flotation machine currently undergoing plant site evaluation, and the HPGR technology now installed in the copper industry. In other cases, the challenge is being met with improved process operation based on circuit simulation, JK Simmet, Modsim, etc., advanced control/instrumentation, and of course, advanced feed characterization with XRF, QEM/SCAN, MLA, etc. Finally, the challenge may be met by improved process strategies such as the first autoclave pressure leaching of copper concentrates as recently developed by FMI to treat 215,000 tons of concentrate per year containing 34% copper.

In China the capacity for alumina production will be nearly 30 million tons per year in 2010 up from 5 million tons in 2003. High quality bauxite resources are already in short supply. Efforts are now being made to develop technology for the processing of low-grade bauxite ores

to economically produce diaspore concentrates for the Bayer process since direct shipping ores are becoming depleted.

In order to meet these challenges imposed by the Tyranny of Ore Grade and Tyranny of Ore Type, we also must prepare clever engineers to serve our profession. This has always been difficult, especially in the U.S. where students are inclined to major in whatever discipline might be most fashionable at the time. In my career, I have seen this with the eras of nuclear, aerospace, computer science, bioengineering, and nanotechnology. Our profession, mineral processing, which dates at least to the beginning of recorded history, is generally not so popular although we believe that a career in mineral processing is a calling of particular significance. Now in the U.S. we have about a dozen universities that offer B.S. degrees either in Metallurgical Engineering or Mining Engineering which require at least one course in Mineral Processing. These programs produce about 100 students each year many of whom do not take employment in the mineral industry. In the U.S. more qualified students are needed to satisfy the demand for employment. Nor are we able to find high quality young faculty to train such students and sustain our Mineral Processing programs. We need to promote our profession and encourage young engineers to take-up careers in mineral processing.

The challenges we face in our profession are significant. While the challenges of resource availability/quality, technology development, and student preparation are great, the opportunities are even greater.

Thank you. It’s wonderful to be with you, and best wishes for a successful conference.

Monday 26 October 2009

Checkout and departure from the Radisson Hotel in Delhi after an excellent breakfast was without incident. The flight to Bhubaneswar from the domestic terminal was a bit tricky since they would not announce the gate until the last minute and I had trouble with the clarity of the communications system and the accent. I was received at the Bhubaneswar airport, not changed too much since 1981, by Ashok Suha and taken to the Mayfair Lagoon Hotel/Resort which is a beautiful place in the setting of a jungle with lovely palm trees and, of course, many lagoons. At the front of the hotel as you arrive by auto is a gigantic elephant statue rising to the second level of the hotel, awesome. The accommodation is great and shortly after arrival I had a swim in a unique swimming pool with most interesting tile design. Before evening I worked on Vishal's paper. In the evening B.K. and Anu Mishra came for me and we had tea and some cashews. No curry except some curried chicken on the flight to Bhubaneswar. Tomorrow morning, Tuesday, I will visit the Institute for Mineral and Materials Technology. B.K. is the Director of this institute. In the afternoon Anu will take me shopping. The International Seminar on Mineral Processing Technology begins on Wednesday morning, Tuesday night in SLC. Here in Bhubaneswar we are about 12 hours ahead of SLC time.

Bhubaneswar, India, Tuesday 27 October 2009

After breakfast at the Mayfair Lagoon, the driver took me to the institute (IMMT) where I met with B.K. and Dr. Biswal for some discussion. Subsequently we went to the Colloids and Materials Chemistry Department for further discussion and to see their laboratories. There we met with Drs. Sarama Bhattacharjee, Bimal Singh, and Laxmidhar Besra. On return to the central office we reviewed my presentation for tomorrow morning and then went for a Chinese lunch at the Mayfair Lagoon. After lunch we went shopping with Anu and her daughter Laurie. We found several outfits and scarfs which we purchased as well as some jewelry from Rajistan province.

After the shopping I returned to the Mayfair Lagoon for reading and a bit of a rest. In the evening we looked for a book before going to dinner. Tomorrow the conference begins.

Bhubaneswar, India, Wednesday, 28 October 2009

We had quite a shopping day on Tuesday with many purchases including some oxidized silver jewelry in the evening before dinner. Dinner was taken at B.K. Mishra's home with his wife Anu, daughter, Laurie, and son Coco. Coco had attended Wasatch Elementary and is about the age of our grandson, Ryan. For dinner we had some “authentic” Indian food, a pizza from Pizza Hut.

Today was the 1st day of the conference, International Seminar on Mineral Processing Technology, at the IMMT campus in Bhubaneswar. After breakfast at the Mayfair Lagoon Hotel/Spa I was driven to the IMMT campus. The auditorium, which holds several hundred people, was packed. I participated in the opening ceremony and gave the remarks I had prepared. Many bouquets of flowers were received which are nicely organized in my room at Mayfair Lagoon. After a break for tea, my Inaugural Lecture, "High Resolution X-ray Micro CT (HRXMT) Advances in 3D Particle Characterization for Mineral Processing Operations" was given. Although this went on for an hour and a half, everyone was quite excited about our work and the lecture was well received. Of course gifts and more flowers followed. Lunch was taken on the campus and then other presentations were made as part of the program. In the evening, at the IMMT campus, we had a cultural event, Utkal Darpan (Mirror of Utkal), a musical presentation of the region's art, including singing and dancing. The conference banquet was then held at the Mayfair Lagoon to conclude the activities on Wednesday.



JDM, A.D. Baijal, and B.K. Misra, International Seminar on Mineral Processing Technology (MPT-2009), Bhubaneswar, October 2009



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Utkal Darpan (Mirror of Utkal) musical presentation, IMMT Campus, Bhubaneswar, October 2009

JDM JOURNAL
New Delhi, INDIA
IMPC 2012

22-29 September 2012

Xuming and I traveled to New Delhi, India, to participate in the International Mineral Processing Congress, IMPC 2012, departing on Delta from SLC on Saturday 22 September 2012, connecting through Cincinnati and then an Air France flight from Paris arriving in New Delhi late Sunday night, 23 September 2012. A driver had been arranged to take us to the Congress Hotel, the Hotel Ashok, and we arrived at about 11 pm. We had papers to present at the congress and I had been invited by the Organizing Committee, Dr. Pradip, and Dr.B.K. Mishra (Raj's former student at the U of U) to present a Plenary Lecture on Thursday. This is the same Dr.B.K. Mishra who arranged for my visit to Bhubaneswar, India, in 2009 for a special mineral processing conference, the International Seminar on Mineral Processing Technology, 26-30 October 2009 (MPT-2009), at IMMT. The IMPC 2012 was held at the Hotel Ashok, the first IMPC in India, and included an exhibition. Our friends from Xradia, S.H. Lau and Sylvia Yun, participated and had an exhibit for their X-ray tomography systems.

Many friends from around the world were present with about 1200 registered for the IMPC, including Janusz and Barbara Laskowski from UBC; Dee Bradshaw from JKMRC; Cyril O'Connor from UCT; Rob and Dee Dunne from Newmont; Nagaraj from CYTEC; Anh and Lin Nguyen from University of Queensland; John and Cindy Herbst from Metso; Raj Rajamani from U of U; many friends from Kunming University of Science and Technology (Professor Dianwen Liu and Dean Shuming Wen); Central South University; BGRIMM, Bejing, China; Jan Drzymala Wroclaw University of Technology, Poland; Richard Williams, University of Birmingham, U.K.; T.C.Rao, formerly Indian School of Mines, India; and K.Moon from KAST in South Korea. The assistant to Dianzuo Wang from the Chinese Academy of Engineering brought a gift for me. It was nice of Dr.Wang to remember me especially since I understand his health is poor. The Fuerstenau brothers (Doug from Berekeley and Maurie from UNR) did not attend because of health problems, in fact Maurie had been in the hospital with pneumonia and passed away unexpectedly on 7 October 2012 from a massive stroke after recovery from the pneumonia. We met Mauricio Bergerman from Universidade Federal de Alfenas, Brazil, who is the professor for Everton and Felipe, students who are with us at the U of U for a training program this semester. Oswaldo Bascur (Osisoft LLC), Amlan Datta, Avimanyu Das, all former students at the U of U attended the IMPC 2012. Also Fernando Concha and his wife (University of Concepcion, Chile) as well as Barry Wills (Editor of Minerals Engineering) were in attendance at IMPC 2012. Others participating included Roe-Hoan Yoon, Virginia Tech, Grahme Jameson, University of New Castle, Australia, Jonas Addai-Mensah from the Ian Wark Center, University of South Australia, Michael Bryson, MINTEK, Republic of South Africa and Peter Scales, University of Melbourne, Australia. Professor Mehmet Celik from Istanbul Technical University was at IMPC with his wife. Many of these people remember my old girlfriend, Patricia, and send their regards.

The first day of the IMPC my watch died and this complicated things a bit since there was no clock in my room. Fortunately, Avimanyu Das, former student, took my old watch and had it resurrected by installation of a new battery. Praise the Lord. I chaired a flotation session on Tuesday while Xuming went with Chinese friends to visit Agra and the Taj Mahal. It was an all day trip to the Taj Mahal over bumpy roads. Xuming was exhausted, returning at about midnight, as I recall. We had an Indian cultural night on Tuesday including Odissi and Dalkhai dance recitals at Hotel Ashok, lots of drums and loud music (?). I did not enjoy it and left early for my room. On Wednesday both Xuming and I had presentations. The presentations went well with many compliments. My Plenary Lecture on Thursday probably had five hundred in attendance. Again many compliments were received and collaboration sought from researchers in South Africa, Bulgaria, Turkey, etc. Xuming's presentation generated considerable interest and discussion of the hydrophobic surface state, a favorite topic of flotation chemistry researchers.



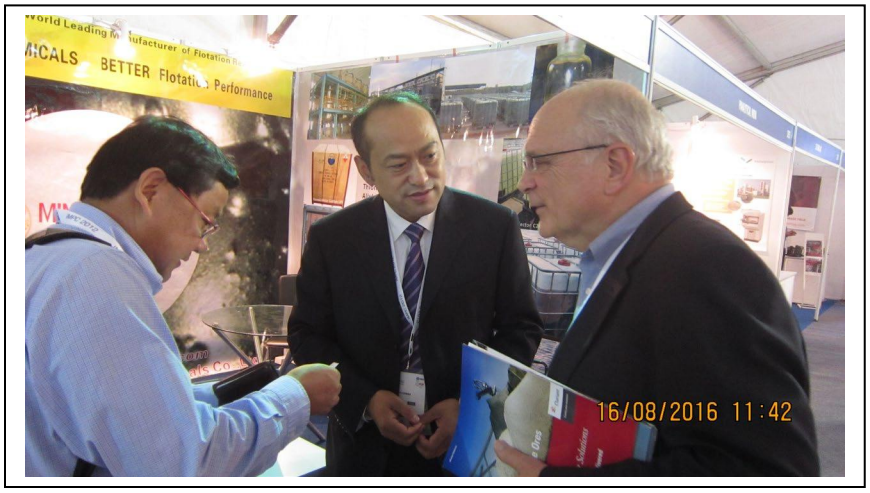
Xuming, JDM, and Raj Rajamani (Metallurgical Engineering Professor and colleague, U of U) at the IMPC 2012 Conference, New Delhi, India.



JDM during his Plenary Lecture at IMPC 2012, New Delhi, India, Thursday, 27 September 2012



JDM during and after his Plenary Lecture at IMPC 2012, New Delhi, India, Thursday, 27 September 2012.



Xuming and JDM at the Shenyang Florrea Chemicals (flotation reagents) booth with Yang Zhiyong at IMPC Exhibition 2012 New Delhi, India.

S.H.Lau and Sylvia Yun from Xradia took Xuming and me for dinner Wednesday night at the Maurya Sheraton Hotel, a 10 minute taxi ride from the Hotel Ashok. Finally on Thursday night we had a show (Samarpan Fusion Band of Bollywood) and the banquet at the India Expo Center and Mart. Driving in New Delhi is exciting and a real adventure. In fact going to the banquet, the bus driver got lost and our arrival was delayed. In order to recover and find his way the driver drove the wrong way on the freeway for about 10 km. I thought that we would have an accident for sure. Talk about an exciting ride! At the banquet Erik Forsberg from Sweden announced that P.C. Kapur had been selected for the Life Time Achievement Award. Kapur did his graduate studies with Doug Fuerstenau at UC Berkeley and had a distinguished career on the faculty at IIT Kanpur.



JDM at the IMPC 2012 Banquet, Thursday evening, 27 September 2012

During the conference, I met Sugandha Sharma who was recommended to me by several faculty from the Indian School of Mines for graduate study at the U of U.

The afternoon of the last day, Friday, 28 September 2012, two ladies from IMMT Bhubaneswar (Swati Mohanti and another lady) took me for shopping in New Delhi at the request of Director B.K. Mishra. Other than the banquet on Thursday night, this was my first day-time excursion from the Hotel Ashok. New Delhi is the capital of India and our driver took us by the government offices, Parliament, the President's home (known as the Brown House in contrast to our White House) and many other buildings. People and traffic were everywhere. It was an exhausting exercise. Nevertheless we found some treasures (eight inlaid white marble coasters) at the Central Cottage Industries Emporium. The marble is that used for the famous Taj Mahal and it is inlaid with coral (red), lapis lazuli (blue), malachite (green), turquoise, and mother-of-pearl. In addition we struck gold at the Tanishq Jewelers and returned satisfied that the results of our expedition would please my girls in the USA.

Finally, before departure on Friday, Professor Subramanian from Bangalore was so kind to present me with gifts, a tie for me and a scarf for my girlfriend. He really wants me to come for an extended stay at his university in Bangalore.

A driver had been arranged by Drs. Pradip and B.K. Mishra to take Xuming and me to the New Delhi airport for a departure shortly after midnight at 12:30 am. We made it through customs/security and flew early morning on Saturday, 29 September 2012, to Paris where the lay-over was 7 hours before we could catch our flight to JFK on Delta. During our trip to the IMPC in New Delhi I thought about Ravi Zacharius, my favorite Christian apologist. New Delhi was the place where Ravi Zacharius lived as a boy and as a young man. I thought about his experiences playing cricket, etc. and this city where Ravi came to Christ in an amazing conversion. See Ravi's book *Traveling from East to West*. Thanks to the Lord for His grace and mercy, the safety of the journey, and another very interesting experience of a different culture in His creation.